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Data Science

VISTA – Vision-Integrated Smart Traffic Automation

Group No. 7

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Project Guide

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Abstract

Traffic signal control (TSC) is a major challenge in urban mobility, where real-time decisions must balance efficiency, safety, and fairness. Conventional systems—fixed-timed, sensor-based, or RL-driven—struggle with dynamic traffic, emergencies, and rule violations. We propose an AI-powered Smart Traffic Automation System that integrates computer vision, deep learning, and language-driven decision-making for adaptive control. YOLO-based perception analyzes live video streams to detect vehicles, pedestrians, emergency vehicles, and congestion for dynamic timing. A large language model (LLM) acts as the decision controller: under normal flow it applies RL optimization to reduce congestion, but in critical cases (emergencies, pedestrian crossings) it overrides with collaborative reasoning to ensure safety and priority. This integration of detection, decision-making, and safety mechanisms provides a scalable, intelligent solution for modern traffic management.

Sustainable Development Goals(SDG) mapped

SDG 11: Sustainable Cities and Communities: Smarter traffic flow reduces congestion and makes cities more livable. Safer pedestrian crossings enhance inclusivity and accessibility. Efficient mobility systems support sustainable urban development.

Introduction

Traffic congestion, frequent rule violations, and delays in emergency response are persistent issues in today's urban environments. Traditional traffic signal systems—either fixed-time or sensor-based—struggle to adapt to dynamic and unpredictable traffic patterns. Even advanced reinforcement learning approaches often overlook critical aspects like pedestrian safety, emergency vehicle priority, and real-time violation detection. These gaps result in inefficiency, longer waiting times, and increased road risks. To address these challenges, there is a growing need for an AI-powered traffic control system that combines the speed of computer vision, the adaptability of reinforcement learning, and the reasoning ability of large language models (LLMs). Such a system should be able to dynamically optimize traffic flow, prioritize emergency vehicles, ensure pedestrian safety, and flag rule violations in real time, all while providing transparency and manual override options for traffic authorities.

Motivation

- **Need for Smarter Control** – Current traffic systems are not fully automated and fail to adapt dynamically, motivating the use of AI-based real-time solutions.
- **Saving Lives** – Emergency vehicles often face long delays; building a system that gives instant green priority can directly support faster response times.
- **Leveraging Modern AI** – Traditional rule-based methods are outdated; integrating computer vision, deep learning, and LLM reasoning offers a chance to create a truly intelligent traffic system.
- **Ensuring Safety & Fairness** – Pedestrian needs and traffic violations are often ignored; automation can improve safety and reduce manual dependency.

Objectives

- To adapt traffic signal timings dynamically using real-time vehicle detection and reinforcement learning, ensuring improved flow and reduced congestion.
- To minimize delays for emergency vehicles by implementing automated recognition and an LLM-guided controller that overrides normal traffic flow during critical situations.
- To enhance pedestrian safety by incorporating YOLO-based pedestrian detection, allowing the system to allocate safe crossing phases for road users.
- To automate traffic violation detection using OpenCV and logging violations for review by authorities.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	[1]A Deep Reinforcement Learning Approach to Traffic Signal Control With Temporal Traffic Pattern Mining	Dongfang Ma Bin Zhou Xiang Song Hanwen Dai	2021	The system uses video feeds converted into grid images and queue statistics as input. A combination of CNN and LSTM processes this data, which is then fed into an A2C reinforcement learning agent to decide whether to keep or change the traffic phase. This approach reduces queue lengths and delays while improving throughput compared to fixed-time and baseline RL methods.	The CNN-LSTM-A2C architecture struggles under heavy congestion, where queue lengths exceed the model's learned thresholds, leading to minimal performance gains. Its reliance on pre-mined temporal patterns makes it fragile against anomalies such as accidents or sudden blockages, as the RL agent cannot adapt to unseen traffic dynamics.
2	[2]A Framework for Adaptive Traffic Signal Control with Deep Reinforcement Learning	Soheil Mohamad Alizadeh Shabestary Baher Abdulhai	2023	The system applies Deep Q-Learning (DQN) to optimize traffic signal phases using raw traffic state inputs such as vehicle counts and queue lengths. The RL agent learns adaptive control policies through simulations, aiming to minimize overall delay and congestion.	The model exhibits instability under non-recurrent traffic events It suffers from overfitting to specific traffic patterns, limiting cross-network generalization. Decisions lack sequential awareness, as the DQN agent does not leverage temporal traffic dynamics.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
3	[3]Intelligent Traffic Signal Automation Based on Computer Vision Techniques Using Deep Learning	Muhammad Talha Ubaid Tanzila Saba Hafiz Umer Draz Amjad Rehman Muhammad Usman Ghani Khan Hoshang Kolivand	2022	The system uses YOLO for real-time object detection on live traffic video feeds to identify and count vehicles at intersections. OpenCV is employed for video frame processing, while a rule-based algorithm adjusts green light durations according to vehicle density. A simple YOLO-based recognition mechanism is also applied to detect emergency vehicles and provide signal overrides.	The system struggles with occlusion and dense traffic. Emergency detection is unreliable, since it relies solely on visual recognition and fails when vehicles are partially blocked. Signal decisions are made frame-by-frame without temporal learning, causing unstable switching under fluctuating traffic conditions.
4	[4]Multi-Agent Transfer Reinforcement Learning for Adaptive Traffic Signal Control	Hongwei Ge Dongwan Gao Liang Sun Yaqing Hou	2022	The framework uses Multi-Agent Reinforcement Learning (MARL) with Deep Q-Networks (DQN), where each intersection acts as an agent. A transfer learning mechanism allows knowledge sharing between agents to accelerate training and improve coordination across multiple intersections.	The approach incurs high communication and training complexity, making coordination difficult. Learned policies have weak generalization when applied to different networks.DQN-based MARL grows complex as intersections increase, leading to slower convergence and unstable policies.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
5	[5]Intelligent Traffic Violation Detection	Roopa Ravish, Shanta Rangaswamy Kausthub Char	2021	The system uses YOLOv3 with Darknet-53 to detect traffic violations at 60 fps. Pre-processed frames undergo smoothing and edge detection, and YOLOv3 identifies vehicles and helmets. It detects helmet absence, seatbelt absence, and red-light violations, integrating results with an E-Challan system for automated fines.	Limited to three violations, requires manual ROI setup, accuracy depends on lighting and camera angle, needs high-end GPUs for real-time performance, and lacks license plate recognition.
6	[6]A Framework for Automatic Detection of Traffic Violations	Aman Malik, Suraksha Bisht, Yukta Tripathi Prashant Kumar, D.R. Gangodkar Nisha Chandran S.	2023	The system uses YOLOv3 with Darknet-53 to detect vehicles in surveillance videos, processing frames via OpenCV and TensorFlow. It flags red-light violations, estimates speed, and enables real-time monitoring through a GUI, with manual ROI setup for infrastructure detection.	The system is limited to red-light violations, lacks number plate recognition and other offense detection, and supports only one video feed at a time. It requires manual ROI setup, has high computational needs, and accuracy decreases with poor ROI visibility or varied signals.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	[7]Traffic Rules Violation Detection using Machine Learning Techniques	P. Srinivas Reddy T. Nishwa R. Shiva Kiran Reddy Ch. Sadviq K. Rithvik	2021	The system applies computer vision and deep learning to detect red-signal violations from CCTV footage. Pre-processing (grayscale, Gaussian blur, background subtraction, thresholding, contour detection) reduces noise, while R-CNN with CNN features and SVM classifies vehicles Violations are flagged when vehicles cross predefined lines during a red signal.	Limited to signal violations, no detection for over speeding/helmet/lane issues. Needs manual line setup and clear footage; accuracy drops in poor lighting or occlusion. Processing is slow, lacks multi-stream support, and license plate recognition is basic.
8	[8]Smart Traffic Violation Detection System using Artificial Intelligence	Kavitha A, Kanimozhi T Dharani J Kishore S Praveen C Vignesh K	2023	CCTV footage is pre-processed using grayscale conversion, Gaussian blurring, background subtraction, binary thresholding, and contour detection to reduce noise. Vehicle detection is carried out with an R-CNN model to classify vehicles. Violations are detected when a vehicle crosses the stop line during a red signal.	he model has high computational cost due to R-CNN, making real-time deployment difficult. Performance reduces in poor lighting and weather conditions. False positives may occur due to occlusion, camera placement, or heavy traffic scenarios.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
9	[9]Emergency Response Optimization System with Dynamic Traffic Signal Control: (Life Saver)	Dr. Rajeshwari B.S Vinay Y Venu Gopal V Tauksik Anil Kumar Viraj Chowdarapu	2024	The system uses an SOS interface to track users' locations and dispatch the nearest ambulance, while an ML-based health interface recommends nearby hospitals. Ambulances are routed efficiently using real-time APIs, and traffic signals are temporarily turned green near approaching ambulances to prioritize their movement.	Drawbacks include infrastructure integration, real-time routing under dynamic conditions, environmental and technical constraints, and limited scalability without cloud and IoT-based signal connectivity.
10	[10]Dynamic Traffic Signal Optimization Congestion Management and Emergency Vehicle Prioritization	Seema U. Deoghare Yash Dhond Saakshi Gaikwad Parth Gorkhe	2025	The system captures real-time traffic images at intersections and sends them to the cloud, where YOLOv8 detects traffic density and emergency vehicles. Signal timings are dynamically adjusted, and neighboring signals coordinate via NRF24L01 modules to ensure smooth traffic flow.	The system has limited scalability for large traffic networks and relies on continuous internet connectivity, making it vulnerable to network disruptions.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
11	[11]Traffic Optimization and Automated Emergency Vehicle Response System	Dr. R. Rajesh Kathir MN Hariprasana Senthilkumar Parthiban SK	2025	The system uses YOLO for real-time vehicle detection and a Reinforcement Learning agent to dynamically adjust traffic signals. The RL agent decides signal changes based on vehicle counts, traffic state, and a reward function that minimizes waiting times and prioritizes emergency vehicles, with dynamic adjustments ensuring fairness and emergency overrides.	The system faces high computational costs and limited scalability, with potential overfitting from simulation-based training. It also encounters challenges in integrating multiple sensors and may be less effective in light traffic or when non-vehicle interference occurs.
12	[12]Smart traffic control system with ambulance detection using image processing.	Dr.M.Nalini S.Sandhya S.Shiwan S.Charumathi G.Maharajothi	2024	The system uses demand-activated traffic lights controlled by a Siemens PLC, with ambulances detected via a YOLOv3-CNN model. Signals turn green for the ambulance and red for all others, resuming normal cycles once it passes.	The system has high initial implementation costs, depends heavily on technology, and involves complex logic that may be challenging to manage across large urban networks.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	[13]Traffic Signal Optimization using Real-Time Pedestrian and Vehicle Counts	Gomathi S Thiruvenkadakrishnan S Sharan Sampath Kumar V B Rithish Kumar G	2025	The system uses real-time video and computer vision to dynamically optimize traffic signals. TensorFlow's SSD MobileNet V2 with OpenCV detects pedestrians and vehicles, while modules for counting and signal optimization adjust green lights based on traffic. A fallback ensures operation during feed failures, with performance measured by wait times, traffic flow, safety, and environmental impact.	The approach is computationally intensive, requiring GPU acceleration for real-time processing. Its accuracy may drop under poor lighting, adverse weather, or heavy traffic occlusion. Scalability remains challenging for large networks, and the system lacks integration of emergency vehicle prioritization.
14	[14]3D-LiDAR-based Pedestrian Detection for Demand-Oriented Traffic Light Control	Dennis Sprute Florian Hufen Tim Westerhold Holger Flatt	2022	Uses 3D-LiDAR sensors and deep learning (ResNet-50) to detect pedestrians and classify vulnerabilities (e.g., elderly, stroller, cyclist). Automates signal requests, reduces waiting time, and adjusts green phase based on needs.	The approach is costly due to 3D-LiDAR sensors, limited to small intersections, and less accurate for thin objects. Performance drops in bad weather, and reliance on a custom dataset makes large-scale deployment difficult.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
15	[15]Advanced Pedestrian Detection Model with Feature Fusion and Imbalance Resolution	R. Karthika R. Kavitha K. Nandhini S. Elakkiya T. Deepa	2025	Combines color, texture, and shape features through fusion and addresses class imbalance using adaptive sampling/weighted loss. Uses deep learning with multi-scale feature maps for better detection of small and occluded pedestrians.	High computational cost, poor performance in low-light or cluttered environments, and challenging real-time implementation due to complexity

Research Gap(Limitations of existing systems)

- Current systems struggle with occlusion, low visibility, and dense traffic scenarios, reducing detection accuracy and reliability in real-world environments.

- There is limited integration of human-in-the-loop oversight, leaving no manual intervention option when AI models fail or behave unexpectedly, which reduces trust and adoption in real deployments.

- Automated traffic violation detection is still underexplored, as most systems rely on manual monitoring or basic sensors, with limited AI-based solutions for detecting red-light jumping, lane misuse, and other rule violations in real time.

- Current research is still heavily dependent on traffic simulations or static sensor data, with very limited exploration of real-time video-based analysis and LLM-driven decision-making for practical, city-level traffic management.

Problem Definition

- Traffic congestion remains a major urban challenge, as existing fixed-time or sensor-based systems fail to adapt dynamically to fluctuating traffic flow, resulting in long delays, fuel wastage, and commuter frustration.

- Emergency vehicles face significant delays because current traffic systems lack intelligent prioritization, preventing ambulances, fire trucks, and police vehicles from reaching their destinations quickly during critical situations.

- Most current systems rely on rule-based or static logic, with limited use of AI-driven decision-making, making them unable to adapt effectively to sudden changes such as road blockages, congestion surges, or incidents.

- Traffic violations such as red-light jumping and lane misuse are still largely monitored manually, making enforcement inefficient and resource-intensive for authorities.

Scope

- Develop an adaptive traffic signal control system that dynamically adjusts signal timings in real time, targeting urban intersections with high congestion to improve traffic flow and reduce delays.

- Implement an automated mechanism to prioritize emergency vehicles with instant green overrides, benefiting ambulance services, fire departments, and law enforcement by ensuring faster response times.

- Integrate pedestrian safety mechanisms to provide secure crossing phases, enhancing road safety for school zones, elderly pedestrians, and crowded market places.

- Deploy an automated violation detection system that identifies red-light jumping and lane misuse, supporting traffic authorities and city administrations in enforcing rules and improving compliance.

Technological Stack

1. Computer Vision and Detection

- **YOLOv11** – Real-time detection of vehicles, pedestrians, and emergency vehicles
- **DeepSORT** – Multi-object tracking for consistent IDs across frames
- **OpenCV** – Video processing, frame extraction, and image handling

2. AI and Decision-Making

- **Reinforcement Learning(PPO)** – Adaptive traffic signal timing optimization
- **LLM (LLaMA Phi-3)** – Context-aware decision support for emergency handling and unpredictable traffic

3. Backend

- **Flask / FastAPI** – API layer connecting detection models with traffic control logic

Technological Stack

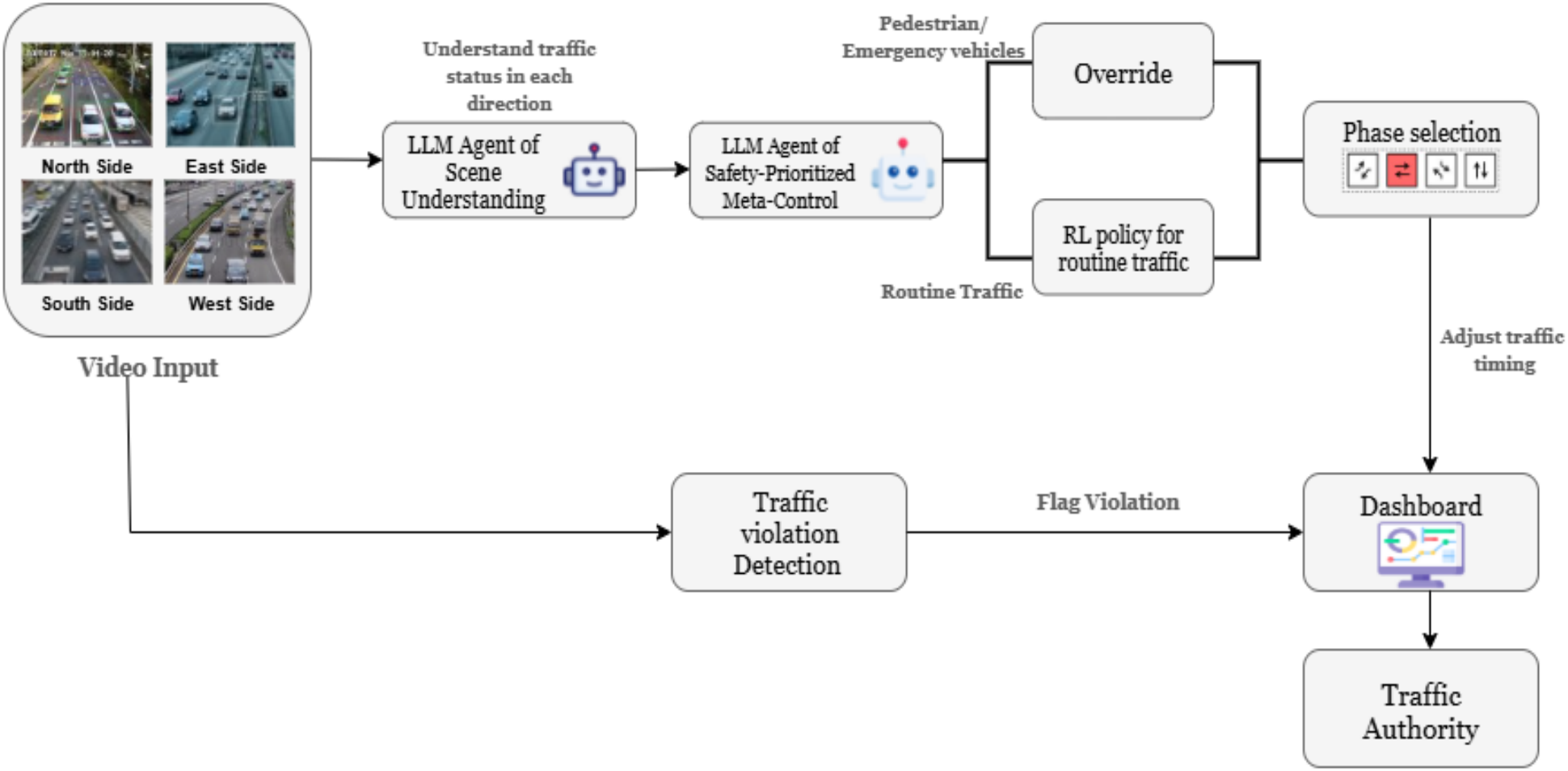
4. Data Processing and Model Training

- **Python (NumPy, Pandas)** – Data handling and preprocessing
- **PyTorch / TensorFlow** – Training and deployment of deep learning and RL models

5. Frontend and Dashboard

- **React.js** – Real-time monitoring and manual override UI for authorities

Proposed system architecture



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Thank You...!!